



Neighborhood analysis with spatial transcriptomics data

The Data Lab

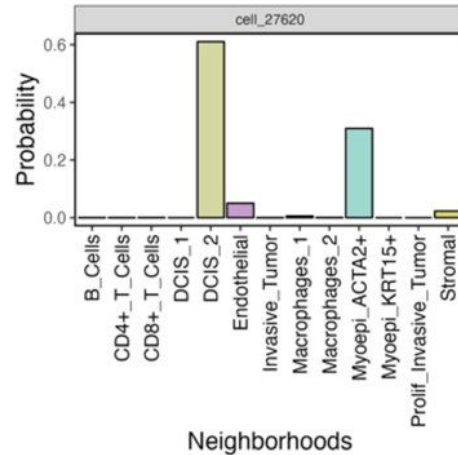
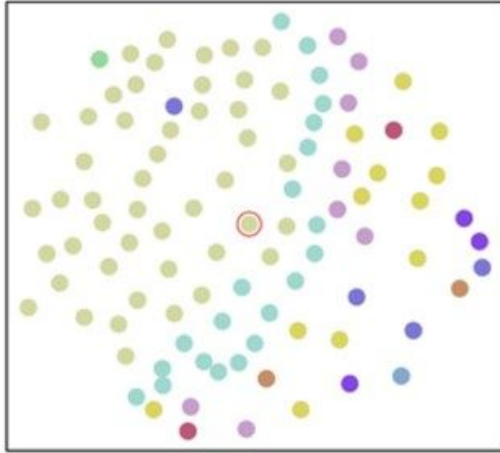
What is cellular neighborhood analysis?

- Method for summarizing which cell types are near other cell types
- A way to use spatial information to identify and characterize the niches or neighborhoods in which cells reside
 - This can be particularly useful for cancer and tissues with complex microenvironments
- Caveat: These tools require spatial data at single-cell resolution
 - Imaging based data (Xenium, CosMx)
 - Transcriptome based data (Visium HD segmented outputs)



Neighborhood analysis outputs a neighborhood profile for each cell

cell_27620



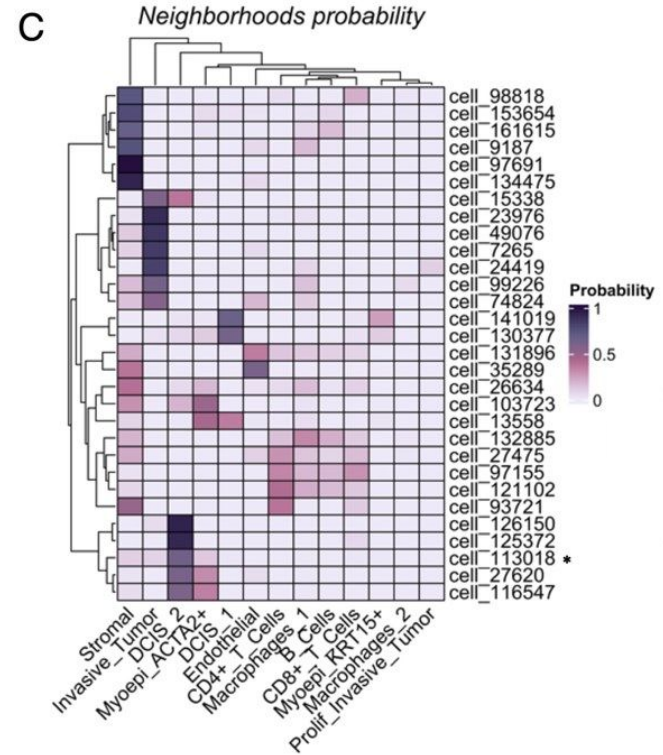
For each cell, calculate the weighted proportion of cells in the local neighborhood corresponding to each cell type.

The **profile** is the vector of these proportions for all possible cell types.

Neighborhood analysis outputs a neighborhood profile for each cell

Neighborhood profiles are combined into a matrix where each row is a cell and each column is a cell type.

But what can you do with these profiles?

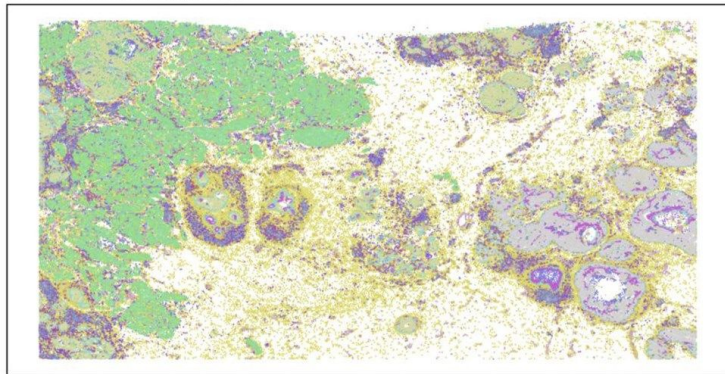


Measure neighborhood complexity

- Perplexity can be calculated for each cell using the identified profile
- This tells us how diverse the local neighborhood is for that cell
- The minimum perplexity is 1, indicating the neighborhood is composed of all the same cell type
- Higher values indicate high complexity

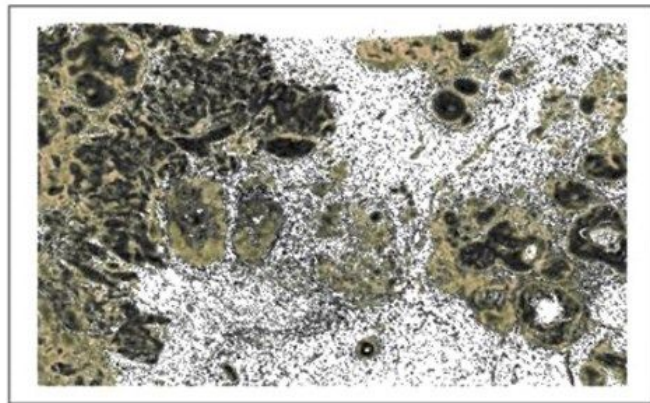
A

Xenium breast cancer sample1 rep1



cell_annotation

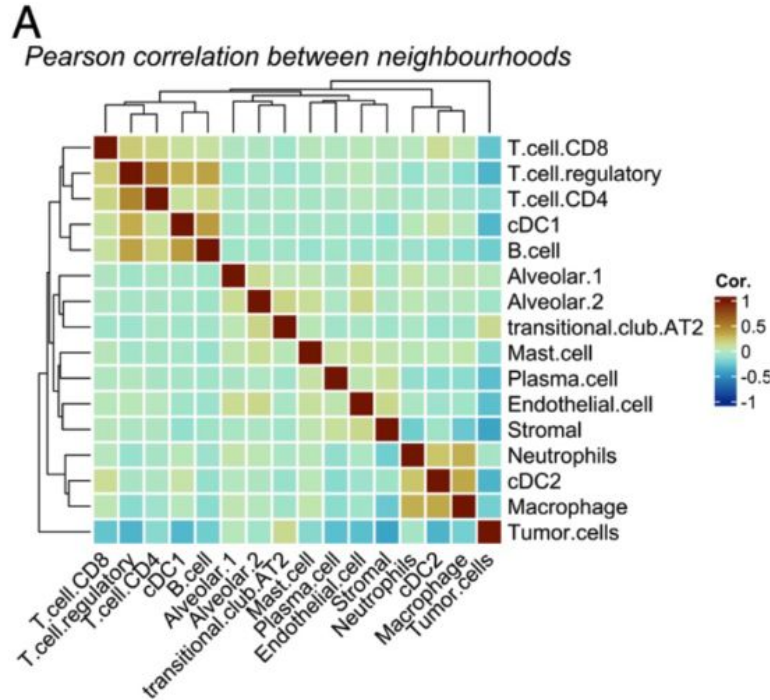
B_Cells	DCIS_1	Invasive_Tumor	Myoepl_ACTA2+	Stromal
CD4+_T_Cells	DCIS_2	Macrophages_1	Myoepl_KRT15+	
CD8+_T_Cells	Endothelial	Macrophages_2	Prolif_Invasive_Tumor	



perplexity

2	4	6	8
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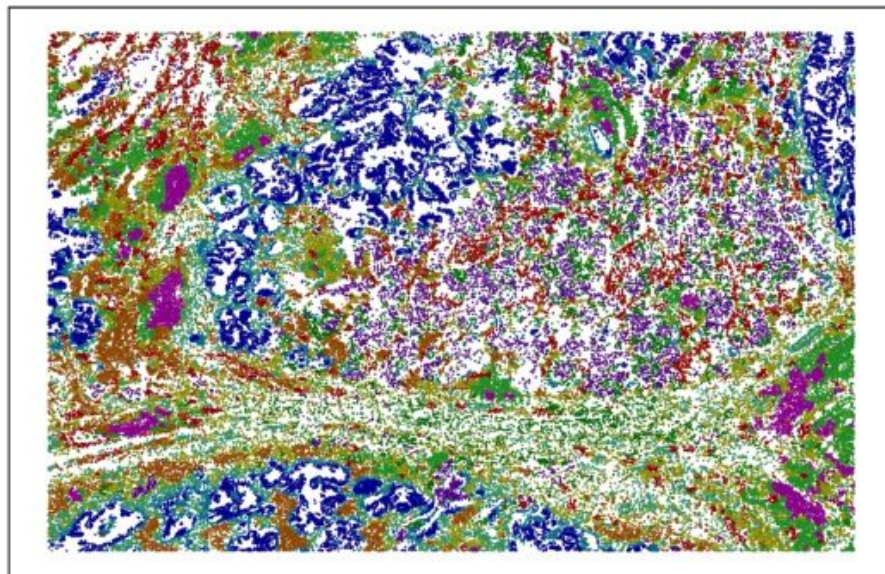
Identify which cell types co-localize



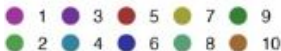
Correlation is calculated between all possible pairs of cells and summarized by cell type

Cluster profiles to identify spatial domains

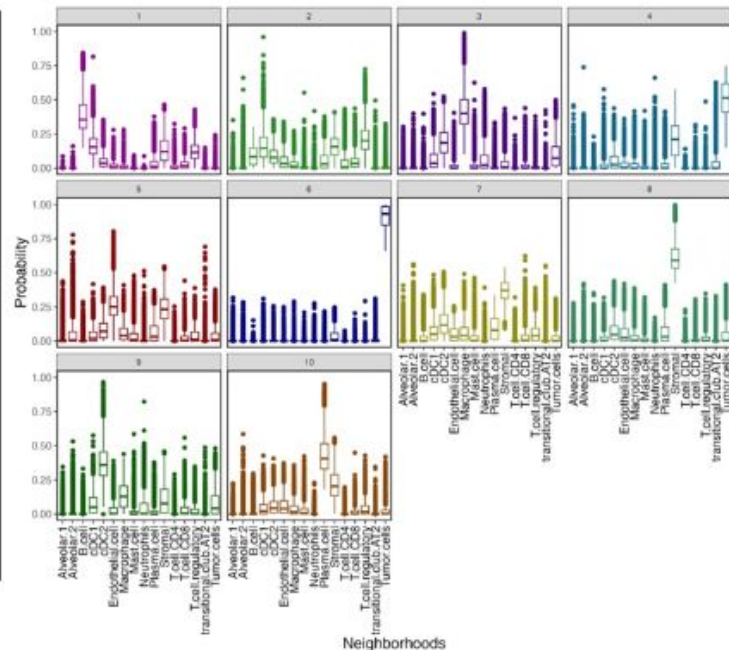
C



Domains



D



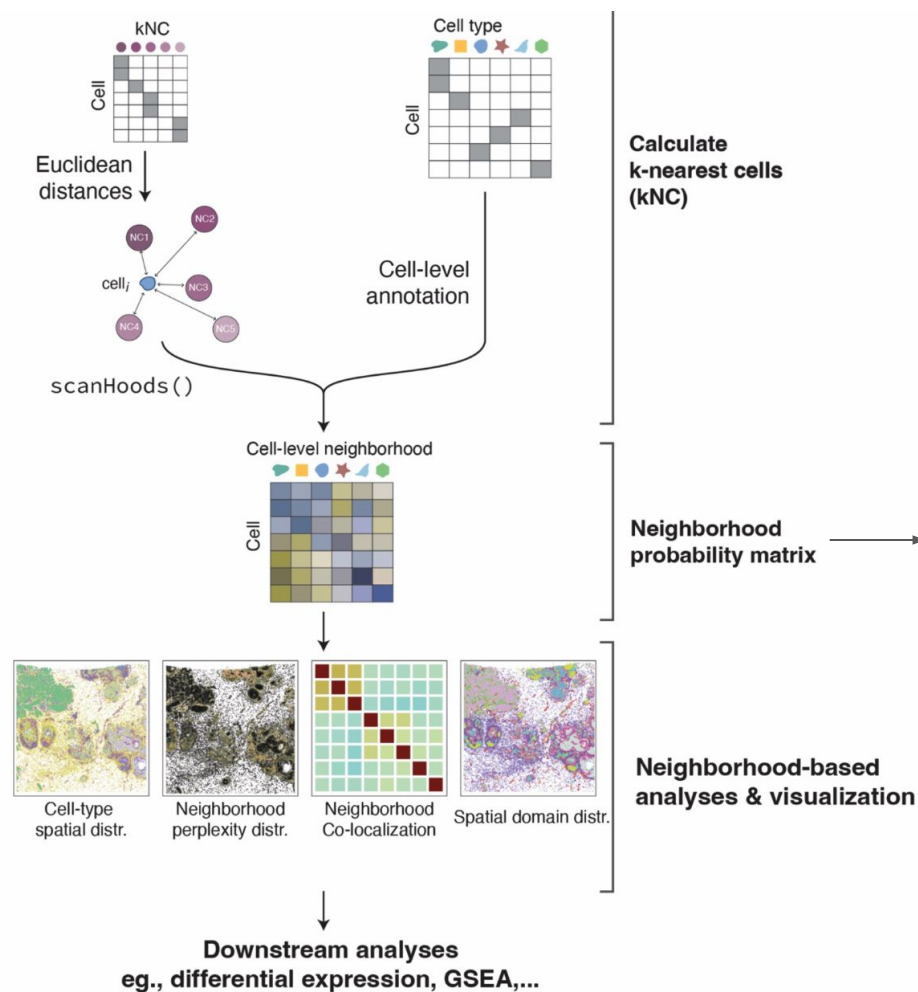
Neighborhoods

Many different packages are available for identifying neighborhood profiles

Features	Giotto(8)	Squidpy(7)	Seurat	Banksy	BayesSpace	MERINGUE	SpaGCN	Stagate	Utag	hoodscanR
Language	R	Python	R	R/Python	R	R	Python	Python	Python	R
Infrastructure	Giotto object	AnnData	Seurat object	Spatial Experiment	Spatial Experiment	NA	AnnData	AnnData	AnnData	Spatial Experiment
Multi-platforms compatible	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Co-localization	Yes	Yes	No	No	No	No	No	No	No	Yes
Multi-neighborhood membership	No	No	No	No	No	No	No	No	No	Yes
Cell-level neighborhood profiles	No	No	No	No	No	No	No	No	No	Yes
Spatial domain detection	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Overall design of hoodscanR

1. Use Approximate Nearest Neighbor (ANN) algorithm to search for the k-nearest neighbors for each cell
2. Calculate a distance matrix between each cell and all neighbors
3. Construct a binary annotation matrix for all cells and all provided annotations
4. Output a probability matrix indicating the percentage of nearest neighbors that have a specific annotation (e.g., probability of finding cell types as a neighbor for each cell)
 - a. Calculated using a SoftMax function with a hyperparameter Tau which controls the shape of the resulting probability distribution and ultimately the influence of neighboring cells

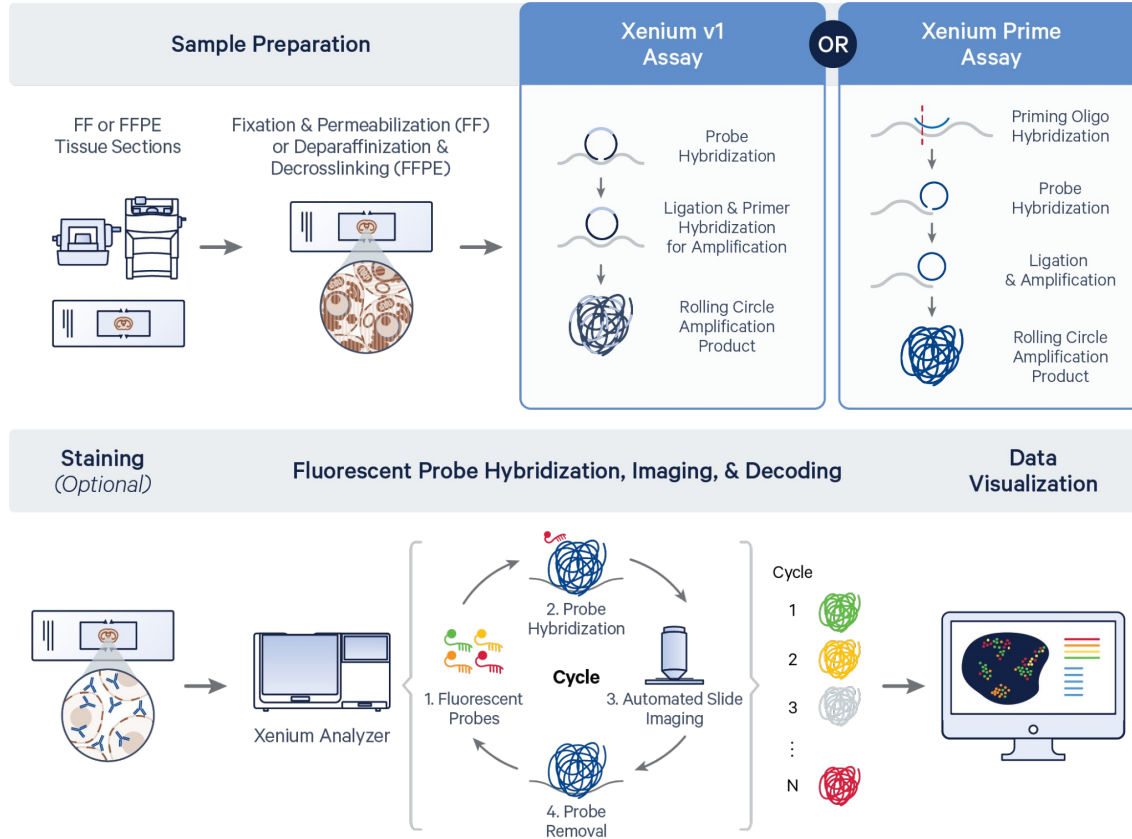


The probability matrix has one row for each cell and one column for each cell type. The values indicate the weighted proportion of cells in the "neighborhood" that correspond to a specific cell type.

Preparing data for hoodscanR

- Spatial platforms that provide cell-level coordinates rather than spot-level coordinates can be used.
 - Imaging based data (Xenium, CosMx)
 - Transcriptome based data (Visium HD segmented outputs)
- Today's instruction will use hoodscanR with Xenium data
- The exercise notebook will use the segmented outputs from Visium HD

Xenium Overview



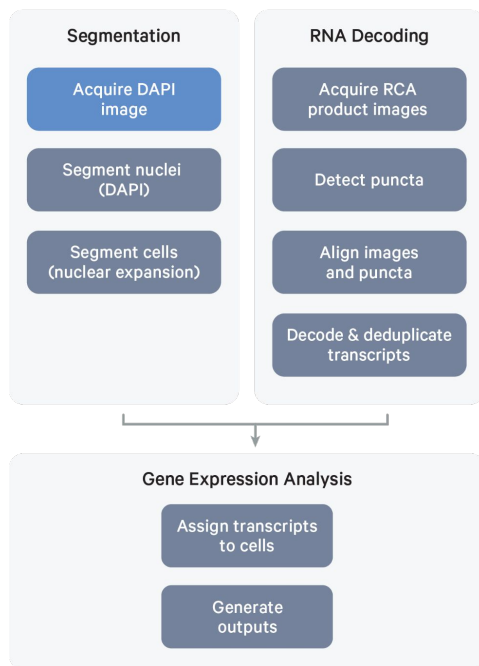
Xenium uses a codebook to turn images into counts

- Each gene is detected by a characteristic pattern of fluorescent signals at a specific site based on hybridization to a gene **probe**
- The pattern is called a **codeword**
- The full lookup table that maps **codewords** to gene probes is the **codebook**
- XOA software decodes the signals and matches it to a **codeword** and calculates a QC score
- Specific **codewords** and reserved **probes** are used as controls to help estimate background and error rates

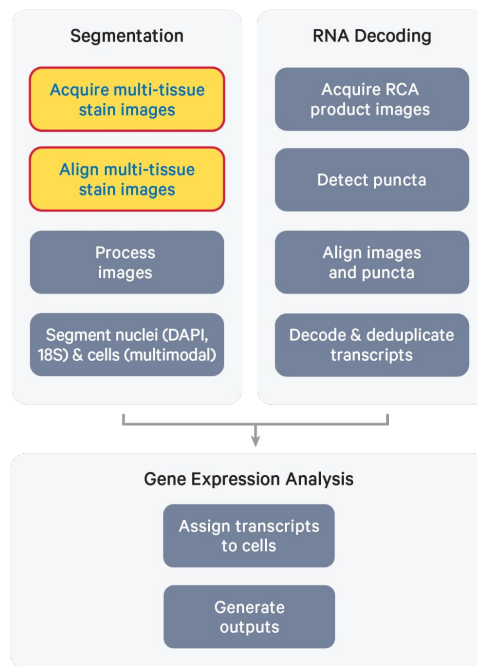
Cell segmentation with Xenium Onboard Analysis (XOA)



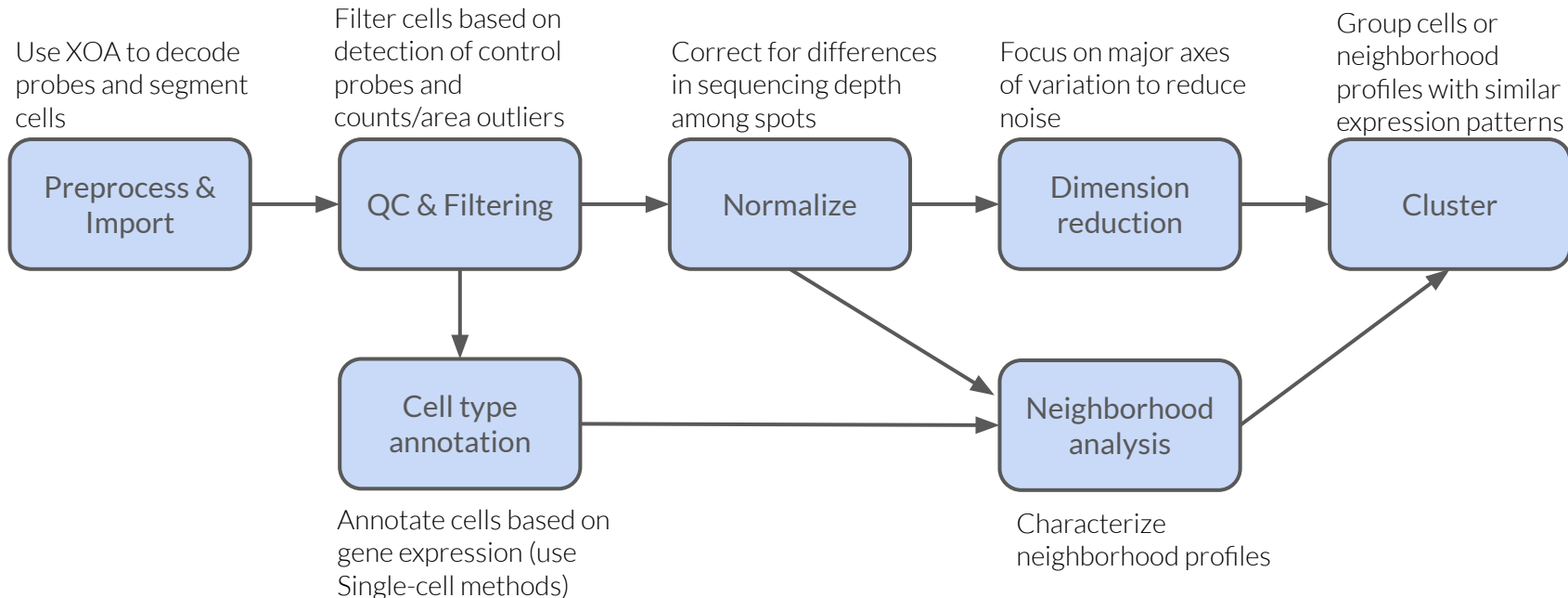
Nuclear expansion segmentation

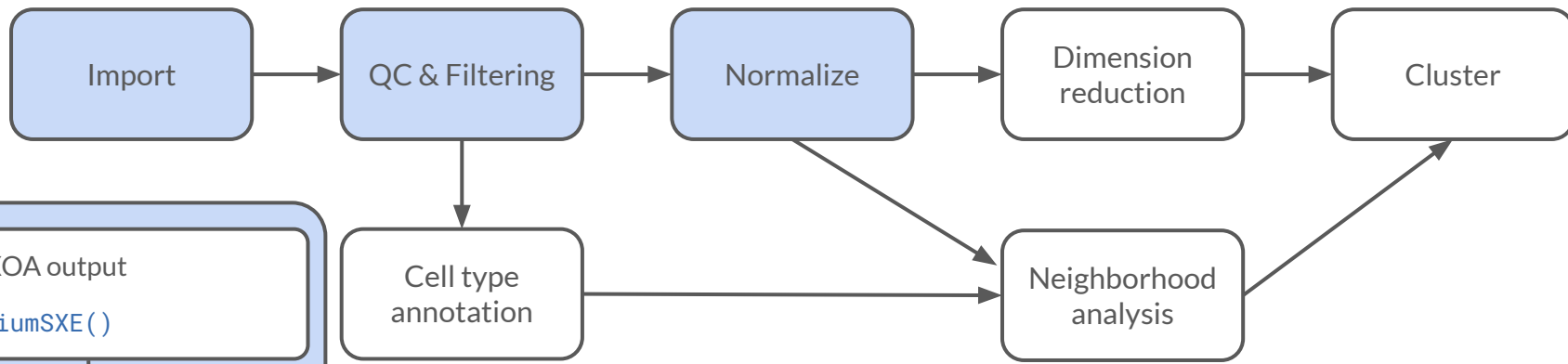


Multimodal cell segmentation



Overview of processing Xenium data



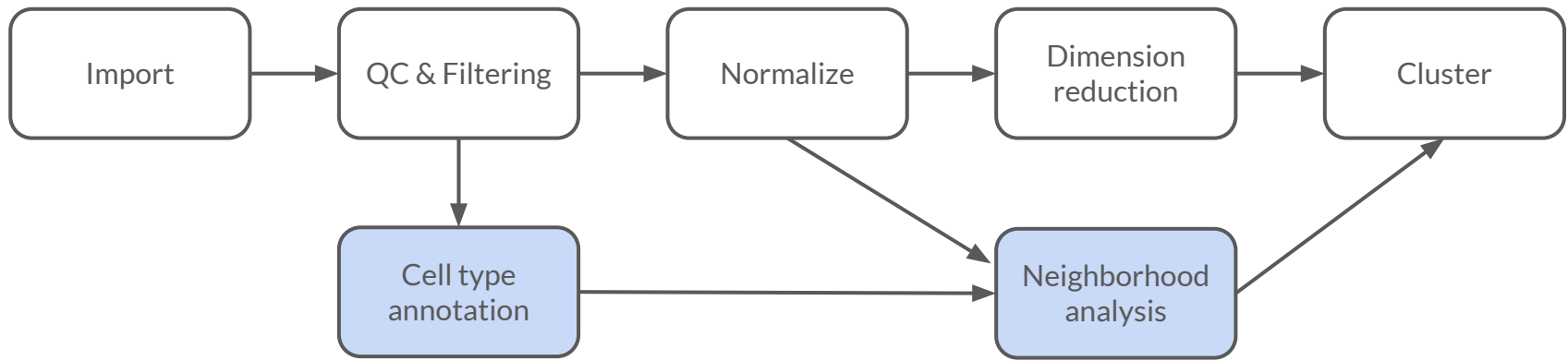


Read in XOA output
`readXeniumSXE()`

Detect low quality spots
`total_counts/cell_area`
`Control_probe_counts > 0 &`
`control_codeword_counts > 0`

Normalize
`computeLibraryFactors()`
`logNormCounts()`

- Outputs from XOA will be directly read into R and saved as a **SpatialExperiment** object
- Two measurements are used to filter out low-quality spots:
 - Detection of control probes or codewords indicate a high level of background noise or errors during the decoding process
 - Cells with a low number of gene expression counts compared to other cells of the same size
- Normalization is the same as with Visium data



- Cell type annotation approaches are the same that are used with single-cell RNA-seq analysis
- **hoodscanR** is used to generate neighborhood profiles for each cell
- Neighborhood profiles are then used for:
 - Co-localization of cell types
 - Calculating neighborhood complexity (perplexity)
 - Clustering into spatial domains